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Group Project

Submission date: **11:00** on **Monday, 7 December 2026**.

> This project must be completed in groups of 3 to 5. Include the full name and SCIPER student number of each group member on the first page of your PDF report; a cover page is not necessary. You must submit your report in PDF format and a separate `zip` file containing your code. Try aim for fewer than 25 pages (all inclusive).
>
> Do not include verbatim listings of code in the PDF report. Your code must run and reproduce the same results as your report. You are free to use "basic" software libraries (such as `numpy`, `scipy`, &c.). Please ensure that figures are exported with good quality and are clear to read in the report.

A note on chatbots from your TA. While a chatbot might be able to complete this whole project, I invite you to see this as a chance to build your understanding and apply some parts of the course which are highly relevant across industry (hey! you could talk about this in an interview).

As your teachers, we already know the answers, and our primary goal is to help you deeply understand the material, rather than see you get to the end as fast as possible. If you have followed the course up until now, then you should find this project challenging but reasonable to complete within the timeframe. Please, do not hesitate to bring questions to the exercise sessions.

If you do talk to chatbots (in addition to, or instead of your TA), you MUST clearly and comprehensively document *where* and *what* the contributions to the project are from the chatbot, and also show where your own contributions can be located in the work. Think about it like this, if you asked an industry expert to complete some parts of this project for you, and then submitted it all as your own work, would that be fair? To be clear, I have not (and will never) include "hidden prompts" within this project to try and trap or trick you. That said, I don't want to get a call one day from the head of risk at some bank or hedge fund, asking me why one of my former students doesn't know how bootstrapping works.

We may invite you to give an oral recitation of your submission if it seems necessary.

— J.H.

Data

Table 1: SOFR quotes.

instrument	quote	maturity
SOFR ON	3.62	20 Aug 2026
SR1 Aug-26	96.361	01 Sep 2026
SR1 Sep-26	96.301	01 Oct 2026
SR3 Sep-26	96.195	16 Dec 2026
SR3 Dec-26	96.072	17 Mar 2027
SR3 Mar-27	96.015	16 Jun 2027
SR3 Jun-27	95.999	15 Sep 2027
OIS 2Y	4.00	21 Aug 2028
OIS 3Y	4.00	20 Aug 2029
OIS 5Y	4.04	19 Aug 2031
OIS 7Y	4.12	19 Aug 2033
OIS 10Y	4.24	19 Aug 2036
OIS 15Y	4.37	19 Aug 2041
OIS 20Y	4.42	20 Aug 2046
OIS 30Y	4.40	21 Aug 2056

Table 1 is the data you will need for this assignment and corresponds to a set of SOFR instruments. The spot date t_0 is 19 Aug 2026; corresponding to trade date 17 Aug 2026.

A reminder about conventions: SOFR futures are quoted as 100 − Rate; for a one-month SOFR futures contract (SR1), Rate is the simple arithmetic average of daily SOFR over that calendar month; for a three-month SOFR futures contract (SR3), Rate is the daily compounded SOFR over the IMM quarter; OIS are annual fixed vs annual compounded SOFR; the day-count convention is ACT/360.[1]

The accrual period of the SR1 Aug-26 futures contract has already started—the spot date lies between 01 Aug 2026 and 01 Sep 2026—so we shall assume that this is a modified futures contract and takes its average from 19 Aug 2026 to 01 Sep 2026. Practically, this means we can infer the 19 Aug 2026 to 01 Sep 2026 forward rate from the SR1 Aug-26 futures rate (under the assumption that the convexity adjustment is zero).

[1] See this CME note for details: SOFR Futures Settlement Calculation.

1 Bootstrapping

Please round all reported figures to four decimal places.

(a) Use the data in Table 1 and the exact bootstrapping method[2] to construct the discount curve $P(t_0, T_i)$ for all reset/payment times T_i corresponding to those market instruments. You may assume that the convexity adjustment between the futures rates and corresponding forward rates is zero.[3]

(b) Reprice (that is, recompute the quote of) each market instrument in Table 1 using your bootstrapped curve and, importantly, utilising the same pricing functionals as you wrote down during the bootstrap. In a table, show the repriced quote as well as the pricing error (in basis points) for each instrument. Interpret your results.

(c) Fill in Table 2: for each given maturity, $P(t_0, T_i)$ is the zero-coupon bond price, $\delta(t_0, T_i)$ is the time-to-maturity under ACT/365, and $y(t_0, T_i)$ is the continuously compounded spot rate (or zero-coupon yield).

Table 2: Zero-coupon bonds.

maturity T_i	$P(t_0, T_i)$	$\delta(t_0, T_i)$	$y(t_0, T_i)$
20 Aug 2026	$x.xxxx$	$x.xxxx$	$x.xxxx\%$
19 Aug 2027	$x.xxxx$	$x.xxxx$	$x.xxxx\%$
21 Aug 2028	$x.xxxx$	$x.xxxx$	$x.xxxx\%$
20 Aug 2029	$x.xxxx$	$x.xxxx$	$x.xxxx\%$
19 Aug 2031	$x.xxxx$	$x.xxxx$	$x.xxxx\%$
19 Aug 2033	$x.xxxx$	$x.xxxx$	$x.xxxx\%$
19 Aug 2036	$x.xxxx$	$x.xxxx$	$x.xxxx\%$
19 Aug 2041	$x.xxxx$	$x.xxxx$	$x.xxxx\%$
20 Aug 2046	$x.xxxx$	$x.xxxx$	$x.xxxx\%$
21 Aug 2056	$x.xxxx$	$x.xxxx$	$x.xxxx\%$

(d) Using Table 2, plot the discount curve; also plot both the spot-rate curve and the (instantaneous) forward curve in the same figure (both these rates should be continuously compounded under ACT/365). For this plot, compute $P(t_0, t_m)$ on a dense grid of daily maturities $t_m = t_0 + m/365$ for $m = 1, \ldots, 365 \times 30$ by using log-linear interpolation on

[2] Slides 116-125 in Part IV: Estimating the Term Structure.
[3] You will compute it later.

the zero-coupon bond prices $P(t_0, T_i)$ in Table 2. Overlay the values $P(t_0, T_i)$ from Table 2 as crosses on top of the interpolated curve.

(e) In Table 2, one of the rows does *not* correspond to the maturity of an instrument in Table 1. Explain which one, how it is calculated step-by-step, and what role it played in the bootstrapping process.

2 Comparing exact and smoothing methods

(a) Using the ten zero-coupon bond prices in Table 2, construct two additional discount curves by calibrating both the Nelson–Siegel (NS) and the Nelson–Siegel–Svensson (NSS) models. Use the sum of squared yield-to-maturity fitting errors as the objective function (slide 138):

$$\arg\min_{\theta} \sum_{i=1}^{10} \frac{1}{10} \big(y(t_0, T_i) - \hat{y}(t_0, T_i)\big)^2$$

where $\hat{y}(t_0, T_i)$ is the yield under the NS or NSS model. Report your calibrated model parameters θ_{NS} and θ_{NSS}, and the square root of the optimal value of the objective function for both NS and NSS in a table.

(b) Use your calibrated NS and NSS curves to reprice (that is, recompute the quote of) each market instrument in Table 1. In a table, under both the NS and NSS curve, show the repriced quote as well as the pricing error (in basis points) for each instrument. Highlight the worst error under each method in bold. Interpret your results.

(c) Plot spot rates and instantaneous forward curves for NS, NSS, and the bootstrap. How do these compare?

(d) Of the three curves you have seen, which one would you use for

- pricing a new instrument?
- computing risk (i.e., duration, convexity, DV01, and PV01)?

Note: This is an open-ended question. We want to see if you can explain a few theoretical and practical considerations for exact vs smoothing methods. A few sentences for each sub-point will be sufficient.

3 Deterministic-shift extensions

In this section[4] you will work with the Hull–White extended Vasicek model under which one has the following short-rate dynamics:

$$dr(t) = \big(b(t) + \beta\, r(t)\big)\, dt + \sigma\, dW^*(t),$$

where W^* is a standard Q-Brownian motion, $\beta < 0$ and $\sigma > 0$ are constants, and $b(t)$ is a deterministic function.

(a) Derive the short rate $(r(t))_{t\geq 0}$. Show your steps.
(b) Show how to define another short-rate process $(\tilde{r}(t))_{t\geq 0}$ such that: $\tilde{r}(t)$ follows a simple Vasicek model, $d\tilde{r}(t) = \beta\, \tilde{r}(t)\, dt + \sigma\, dW^*(t)$, with $\tilde{r}(0) = r(0)$, and where $\phi(t) = r(t) - \tilde{r}(t)$ is a deterministic function with $\phi(0) = 0$ that is one-to-one related to $b(t) = \phi'(t) - \beta\, \phi(t)$.
(c) Derive the bond price in the form $P(t,T) = \mathrm{e}^{-\int_t^T \phi(s)\, ds}\, \tilde{P}(t,T)$ where $\tilde{P}(t,T)$ is the bond price of the Vasicek model.
(d) Let $s \mapsto f_0(s)$ be the initial forward curve. Prove that the Hull–White model fits the initial forward curve f_0 if and only if $\phi(s) = f_0(s) - \tilde{f}(0,s)$ for all $s \geq 0$, where $s \mapsto \tilde{f}(0,s)$ is the forward curve of the Vasicek model.

4 Calibrating a short-rate model and pricing a cap

In this part, you will use the zero-coupon discount curve from Table 2 in Section 1, and the Hull–White extended Vasicek model from Section 3.

For the initial forward curve f_0, use the instantaneous forward curve (continuously compounded with ACT/365) which you derived using the zero-coupon bond prices in Table 2 together with log-linear interpolation.

For the Hull–White model, set $\sigma = 0.01$ (100 bp) and $\beta = -0.25$.

(a) Explain how to set $\tilde{r}(0)$ so that $\phi(0) = 0$ and compute this value. (*Hint*: What is $f_0(0+)$?)
(b) Explain the sense in which the fit to the initial (forward) curve is *exact*. You should explain this theoretically in a few sentences and also show it numerically.
(c) Briefly explain what would change if you used the NS or NSS curve from Section 2 for the initial forward curve.

[4] This section requires some mathematical derivations — show your work!

(d) Derive and compute the convexity adjustment for each SR3 futures contract in Table 1 under the Hull–White model. Show your results in a table, and comment on whether neglecting the convexity adjustment was justified in Section 1.

Next, let us price a five-year quarterly cap with at-the-money forward strike. Here we can write the spot date as $t_0 = 0$, the first reset date is $T_0 = 1/4$, and $T_i - T_{i-1} = 1/4$ for all $i = 1, \dots, 19$, where $T_{19} = 5$ is the maturity.

Consider both cases of a cap on forward-looking rates ($LR = L$) and on backward-looking rates ($LR = R$).

(e) Calculate the at-the-money strike κ. Explain why κ is the same for forward-looking rates ($LR = L$) and backward-looking rates ($LR = R$).
(f) Under the relevant forward measure, show how to convert the price of each underlying caplet to the price of bond options.
(g) Using the zero-coupon bond prices from your bootstrapped and interpolated zero curve, compute the price of each caplet as well as the cap price. Quote the price in terms of basis points per unit notional.
(h) Compute the Black implied volatility for each caplet and for the cap.
(i) Explain the shape of the caplet by caplet implied volatilities. Explain how to interpret these and how they relate to σ?